



Gold Price Prediction and Market Analysis

Machine Learning Internship Project Report

Executive Summary

This project analyzes 10 years of historical gold prices to develop predictive models, formulate algorithmic trading strategies, and evaluate market sentiment during global crises.

Objective 1 & 2: Predictive Modeling (ARIMA vs. Prophet)

- **Initial Approach:** We initially tested an ARIMA(0,1,1) model. However, predicting a full year at once resulted in a flat line, as the Random Walk nature of the model reverted to the mean.
- **Advanced Optimization:** We pivoted to Facebook Prophet to better capture the strong yearly seasonality and long-term trend. By implementing Multiplicative Seasonality, adding US Holidays, and increasing trend flexibility, we achieved a highly accurate forecast.
- **Results:** The final Prophet model achieved a **MAPE of 6.88%** and an **RMSE of \$149.86**, meaning the model's predictions were over 93% accurate on the hidden test data.

Objective 3: Trading Strategy Backtesting

We developed an algorithmic trading bot using a Moving Average Crossover strategy (20-Day Fast SMA vs. 50-Day Slow SMA) with a starting capital of \$10,000.

- **Normal Strategy (Trend Following):** Generated a profit of **\$2,329.50 (23.29% ROI)** by riding the momentum of the 10-year bull market.
- **Reverse Strategy (Contrarian):** Generated a profit of only **\$1,232.50 (12.32% ROI)**.
- **Conclusion:** Reverse trading is not optimal for profit maximization in a strong long-term bull market, as it constantly fights the prevailing upward trend.

Objective 4: Market Sentiment Analysis

Gold's status as a "Safe Haven" asset was visually and statistically confirmed.

- During major global crises (COVID-19 in Feb 2020, Russia-Ukraine in Feb 2022), the data showed massive spikes in trading volume representing market anxiety.
- Immediately following these fear-driven events, gold experienced sharp upward price momentum as investors fled risky stock markets for the safety of gold.

Objective 5: Statistical & Risk Analysis

- **Volatility:** Gold exhibits an average daily volatility (standard deviation) of **0.93%**.
- **Kurtosis (3.81):** The daily returns distribution exhibits "fat tails." This indicates that extreme price shocks happen more frequently than a normal Gaussian distribution would predict.
- **Correlation:** Trading Volume showed almost zero linear correlation (0.02) with the Closing Price, proving that high volume indicates market activity/panic rather than directional price movement.